



DISSERTATION

Making Differentiable Neural-Architecture Searches Applicable and Hardware-Aware

to obtain the academic degree

DOKTOR-INGENIEUR

(Dr.-Ing.)

at the

Faculty of Engineering, Computer Science and Psychology
of **Ulm University**

by

Joschua Conrad

from Marburg

Supervisor: Prof. Dr. Maurits Ortmanns

Second Reviewer: Prof. Dr. Vasileios Belagiannis

Third Reviewer: Prof. Dr. Walter Karlen

Officiating Dean: Prof. Dr. Matthias Tichy

Ulm, May 6, 2026